practices, I encourage you to head over to the Google SRE books site (https://sre.google/books/) and read everything. In this chapter, we will take a more practical approach, covering the fundamentals quickly and then showing how to use them.

The main objective is to make the availability of a service measurable and compare the metrics derived from services with goals. Once you have the goals agreed upon and you are in a position to automatically measure them, you can do a lot more:

- You can draw nice charts and look at them, which is good for usage in frontends.
- You can alert if you're at risk of breaching a goal (see also our conversations in chapter 7 on cloud operations).
- You can use these metrics as an input in a contract with your customer to promise something, and in case you are not able to deliver what you promised, you can commit to a financial compensation (possibly credits or even penalty payments). This means you're earning trust from your customer in a quantifiable and systematic manner.
- You can drive compliance (e.g., in the financial or health care domain).

> **Note**
>
> There is a parallel to the ideas we discuss in this chapter in real life: when one of our kids and I agree there will be a reward for, say, a clean room, we first need to define and agree on what *clean* means (that would be the contractual piece). I might say, for example, that over a month's time, I expect to only find three pieces of dirty clothes on the floor (that's the goal the child needs to reach to qualify for the reward). Then, I would randomly inspect the room and note how many dirty clothes I find on the floor (the measurement) and, at the end of the month, compare with the goal. If I were to find three or fewer pieces of clothes on the floor, I would declare the goal successfully completed, and the child would get their reward.
>
> At any point in time, the child could look at my list and see how many pieces of clothes (in the measurement period here, which is one month) I found. With that, they could determine how close they were to breaching the agreed-upon goal or, in other words, how much wiggle room they had left (you could call that a *budget*, and indeed, that is the official term for it: an *error budget*).

In this chapter, we will first tackle service level objectives on a theoretical level; however, I'm afraid there is a bit of terminology we need to untangle first. Then, we will shift gears and have a look at tooling available in this space, and you will complete a hands-on session of implementing availability metrics and comparing with the goals you defined. In the last section, we will discuss some important considerations and provide further reading.

## 10.1 The fundamentals of SLOs

In this chapter, we differentiate between three similar-sounding but entirely different terms: service level agreements (SLAs), service level objectives (SLOs), and service